KOLEJ UNIVERSITI TUNKU ABDUL RAHMAN

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2020/2021

DECEMBER EXAMINATION

COMPUTER SCIENCE BACS2163
SOFTWARE ENGINEERING

MONDAY, 14 DECEMBER 2020 TIME: 9.00 AM – 12.00 NOON (3 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS) IN INTERACTIVE SOFTWARE TECHNOLOGY
BACHELOR OF COMPUTER SCIENCE (HONOURS) IN SOFTWARE ENGINEERING

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS DEVELOPMENT

Instructions to Candidates:

Answer **ALL** questions in the requested format or template provided.

- This is an open book final online assessment. You MUST answer the assessment questions on your own without any assistance from other persons.
- You must submit your answers within the following time frame allowed for this online assessment:
  - The deadline for the submission of your answers is **half an hour** from the end time of this online assessment.
- Penalty as below **WILL BE IMPOSED** on students who submit their answers late as follows:
  - The final marks of this online assessment will be reduced by 10 marks for answer scripts that are submitted within 30 minutes after the deadline for the submission of answers for this online assessment.
  - The final marks of this online assessment will be downgraded to zero (0) mark for any answer scripts that are submitted after one hour from the end time of this online assessment.
- Extenuation Mitigating Circumstance (EMC) encountered, if any, must be submitted to the Faculty/Branch/Centre within 48 hours after the date of this online assessment. All EMC applications must be supported with valid reasons and evidence. The UC EMC Guidelines apply.

FOCS Additional Instructions to Candidates:

- Include your **FULL NAME**, **STUDENT ID** and **PROGRAMME OF STUDY** in your submission of answer.
- Read all the questions carefully and understand what you are being asked to answer.
- Marks are awarded for your own (original) analysis. Therefore, use the time and information to build well-constructed answers.

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STUDENT'S DECLARATION OF ORIGINALITY
By submitting this online assessment, I declare that this submitted work is free from all forms of plagiarism and for all intents and purposes is my own properly derived work. I understand that I have to bear the consequences if I fail to do so.

Final Online Assessment Submission
Course Code:
Course Title:
Signature:
Name of Student:
Student ID:
Date:

BACS2163 SOFTWARE ENGINEERING

Question 1

As of March 2020, the education sector around the world is confronting a gigantic effect because of COVID-19, with universities compelled to move their education to e-learning and mixed learning modes. TAR UC academics and students have needed to adjust to this change, both by the method of instructing, and learning procedure. As TAR UC sets to keep lectures online until the end of the year (December 2020), the management needs a new system as soon as possible to ease the teaching and learning experience for their academics and students.

You are the project manager and is responsible to build the new teaching and learning system for TAR UC. The main modules to be developed are still not finalised so you need to interview the academics and the students to get the user requirements and you may expect that the user requirements could change frequently. The management has set a low budget for this new system thus you believe that the new system is not that complex. You plan to allocate only one junior software engineer and one intern staff in getting the system to be ready as soon as possible.

a) Analyse if the new system as mentioned above is a generic software product or bespoke software product? Explain your answer with **TWO (2)** reasons. (1 + 4 marks)

b) Select and explain **ONE (1)** software process model to develop the new system as mentioned above. Organise your selection with **FOUR (4)** reasons. (2 + 8 marks)

c) Identify and explain **THREE (3)** software quality attributes for the new system as mentioned above. Justify your answer. (6 marks)

d) As a project manager, you would need to consider the factors that could motivate the new intern staff throughout the project. Identify and explain **TWO (2)** motivating factors for this case based on the Maslow Motivation Model. (4 marks)

[Total: 25 marks]

Question 2

Your uncle owns a shipping company and he wants to computerise a series of services such as vessel booking, rental and return of the vessel, rental payment, manage account et cetera. Customers will be given an online login ID with a password to ease the entire booking process. You are in charge of the project.

a) Illustrate a Gantt Chart for the project based on the following table. Assume that the project starts in October 2020. (9 marks)

Task ID	Task Description	Predecessor	Duration (month)	Overlap (month)
A	User Requirements	None	½	None
B	System Analysis & Design	A	½	¼
C	End-user Feedback	A	½	¼
D	Coding	B, C	1	¼
E	End-user Feedback	B, C	1	¼
F	Testing	E	¼	None
G	Deployment	F	¼	None

Question 2 (Continued)

b) Identify **FIVE (5)** functional requirements and **TWO (2)** non-functional requirements for the project. (5 + 4 marks)

c) Identify **FIVE (5)** viewpoints for the project by illustrating a viewpoint hierarchy to present your viewpoints. (7 marks)

[Total: 25 marks]

Question 3

Nozomi Logistics and Express Sdn. Bhd. is a Malaysian based company who provides reliable, efficient and courteous transportation services with over 15 years of experiences in the related industries. Nozomi has branches in Johor Bahru, Melaka, Butterworth, and Singapore. The president of the company is planning to computerized a series of services such as consultation, warehousing, freight forwarding, delivery services, and customer support in their daily working environment.

a) Identify and explain **TWO (2)** testing techniques that can be used in testing Nozomi computerized system. Generate **TWO (2)** test cases for each identified testing technique above to assist the testing process by using the following format. (5 + 4 marks)

Program Name: Test Date: -					
No.	Test Case	Test Data	Expected Results	Actual Results	Remarks
				-	-
				-	-

b) Recommend and explain **TWO (2)** user interface design principles that must be applied in the proposed Nozomi computerized system as mentioned above. Justify your recommendation. (5 marks)

c) Suggest and explain **TWO (2)** functional classification of Computer Assisted Software Engineering (CASE) tools with **ONE (1)** specific example of tool for each classification that can be used during the software development project of Nozomi computerized system. (4 marks)

d) Identify and construct a diagram to show **THREE (3)** core concerns and **THREE (3)** cross cutting concerns for Nozomi computerized system. (7 marks)

[Total: 25 marks]

Question 4

a) You, a software engineer is currently designing a real-time multitasking humanoid sized robot which can assist the elderly who lost their autonomy to open doors, climb stairs, and reach objects while going about its care duties.

(i) Construct **TWO (2)** sets of aperiodic stimuli-response and **TWO (2)** sets of periodic stimuli for the mentioned real time multitasking robot system. (6 marks)

(ii) Differentiate the best and the worst cohesion level that has been suggested by Constantine & Yourdon (1979). Select and justify **ONE (1)** level of cohesion that is suitable for this system. (2 + 2 marks)

b) Laplaya Restaurant has a legacy system that stores the restaurant transactions, staff, customer, supplier, and ingredients stock information since the 1990s. The system is getting outdated and unable to support new technologies such as wireless ordering, online transaction, and electronic payment in managing their well growing business. Mr.Doni the owner of the restaurant recently contacted you as a software engineer to solve this issue.

(i) Laplaya Restaurant legacy system possessed the following characteristics:

- high dependency between one module to another
- developed by low-level programming language
- tested for 3 months before deployed
- all versions are stored in a configuration management system
- developed in 1990

Analyse each characteristic of the legacy system and comment how it can affect the maintenance cost of the system. (10 marks)

(ii) Assess Laplaya Restaurant legacy system from the aspect of business value and system quality. Suggest and explain **ONE (1)** legacy system management strategy to Laplaya Restaurant. (2 + 3 marks)

[Total: 25 marks]